

Human-Machine Friendly Jing-Zhang

Shared movement, work and governance in the embodied-intelligence era

Credit · Dr. Future wepon · GitHub weponusa

Human first · inclusive access · intelligible interaction · opt-out and accountable · data minimization

Human-Machine Friendly Jing-Zhang

EMBODIED-INTELLIGENCE SHARED CORRIDOR

01 / OVERALL CONCEPT

TWO FOUNDATIONS

- Physical: universal access · all-age walking/cycling · fully human-operated alternative
- Digital: permission scheduling · failure takeover · aggregated performance · auditable records

SHARED SPINE

- Human priority · machines enter conditionally
- Zhongzhiyuan: closed testing / safety validation
- AI Origin Community: public co-creation / prototype incubation
- Dazhongsi: accessible interchange / real-world validation

FIVE SCENES

- Movement — human-priority sharing
- Work — safe collaboration
- Service — human alternative
- Social life — opt-out interaction
- Governance — accountability remains human

FIVE FRIENDLINESS DIMENSIONS

- Mobility — continuous access
- Work — clear duties
- Environment — low disturbance / low carbon
- Interaction — transparent and intelligible
- Governance — audit and accountability

NON-NEGOTIABLE PRINCIPLES

- Human first · inclusive accessibility · transparent and understandable
- Opt-out and accountable · data minimization

From “Machine Entry” to Human-Machine Friendliness

02 / DIAGNOSIS

QUANTITATIVE STRUCTURE + RIGHTS / SAFETY / GOVERNANCE GAPS

THREE SCALES • Coordinated study 43.6 km² • Overall design 11.4 km² • Key areas 368.4 ha · 32.3%	LINEAR FORM • North-south ≈9.7 km • East-west ≈1.4 km • Aspect ratio ≈7:1	THREE-AREA STRUCTURE • Zhongzhiyuan 52% • AI Origin Community 28% • Dazhongsi 20%
P1-P3 SPATIAL / INDUSTRIAL GAPS • P1 Excessive linearity — no baseline for accessible east-west stitching • P2 Broken industry chain — no shared pilot platform from R&D to testing to commercialization • P3 Decorative greenbelt — abundant green space but insufficient operation and services	P4-P6 RIGHTS / GOVERNANCE GAPS • P4 Sharing is not friendliness — no rules for identity, intent, exit or takeover • P5 Vulnerable groups absent — average users conceal all-age differences • P6 Governance/data vacuum — no baseline for incidents, near misses, complaints or privacy	CORE FINDING • The question is not whether robots exist, but whether people become safer, freer and more equal—and whether machines are controllable, intelligible and accountable.

Human-Machine-Friendly Roles of the Three Key Areas

03 / KEY AREAS

CLOSED TEST → CO-CREATION → PUBLIC VALIDATION

ZHONGZHIYUAN · TECHNOLOGY / HARDWARE VALIDATION

- Closed test loop · safety protocols · sensor calibration
- No high-risk testing in public space
- Pre-entry inspection, insurance and degraded-mode testing
- Public observation window + H0 quiet boundary

AI ORIGIN COMMUNITY · R&D / PROTOTYPE INCUBATION

- Co-creation lab · heritage nodes · community life
- Older people, children and disabled people join usability tests
- Machine identity and intent must be visible
- Staffed service window + sensing-free route

DAZHONGSI · PRODUCT / COMMERCIAL VALIDATION

- Accessible interchange · business lounge · user validation
- Public experience before product display
- Refusing smart service must not reduce service quality
- Separated waiting + human takeover

CROSS-CUTTING SYSTEMS

- Space: H0 quiet / H1 slow mobility / H2 sharing / H3 interchange
- Interaction: multimodal cues · human alternative · opt-out
- Safety: fencing · emergency stop · takeover · failure isolation
- Governance: accountable-party registry · data minimization · complaint review
- Industry: foundation → pilot → commercialization → operational feedback

04 / MOBILITY + BLUE-GREEN

H0-H3 RIGHT-OF-WAY + UNIVERSAL ACCESS + SAFE FALLBACK

H2 PRINCIPLE SECTION · ORDER, NOT SURVEYED WIDTH

- Ecological buffer — no entry into sensitive zones
- Continuous walking/accessibility — minimum clear width first
- Interaction buffer — separation between people and machines
- Low-speed machine sharing — identity / intent / emergency stop
- Transfer bay — never occupies the main movement surface
- Shade / stormwater / rest — all-age seating

ORDER CANNOT BE REVERSED

- Guarantee continuous walking and accessibility before allocating machine space.
- Where existing width is insufficient, reduce machine space—not pedestrian space.

CASE-REFERENCE MINIMA • FIELD CHECK REQUIRED

- Clear width ≥ 2.5 m
- Turning radius ≥ 0.8 m
- Slope $\leq 1:12$
- Machine speed in pedestrian space ≤ 1.2 m/s
- Main entrances step-free

H0-H3 RIGHTS

- H0 Human quiet zone — heritage / children / rest; autonomous machines prohibited
- H1 Slow-mobility priority — absolute human priority; approved service machines enter briefly
- H2 Low-speed sharing — time/zone control, intent cues, passing bays, emergency takeover
- H3 Perimeter interchange — accessible station-city transfer; no crossing heritage core

SIX SPACE MODULES

- 01 Human-only quiet · 02 All-age rest · 03 Passing/avoidance
- 04 Transfer bay · 05 Maintenance/charging · 06 Human takeover

SPEED STATEMENT

- H2/H3 speed depends on equipment, spatial risk, field testing and approval; no uniform 20 km/h is assumed.

Linkage Intensity × Human-Machine-Friendly Performance

03 / METRICS + EVIDENCE

KNOWN SPATIAL EVIDENCE + UNKNOWN BASELINES + AUDITABLE GOVERNANCE

KNOWN / RECOMPUTABLE SPATIAL EVIDENCE

- Site area 11.4 km²
- Green-space ratio 36.5%
- Public-space ratio 13.5%
- Road framework 41.5 km
- Scenario nodes / cards 12 / 13

LIFECYCLE GATES

- Before entry: inspection / insurance / impact assessment
- During operation: fencing / cues / logs / emergency stop
- During anomaly: protect people first / safe degradation
- After operation: review / disclosure / downgrade and exit

INDUSTRY LINKAGE · TARGET FRAMEWORK, NOT MEASURED

- Spatial · technological · industrial · talent · scenario connections

EVIDENCE RULE

- Spatial areas are recomputable. HMF baselines must be established first; no fake radar values or targets presented as current performance.

HMF METRICS · ALL BASELINES UNKNOWN

- Mobility: continuous accessibility / clear-width encroachment
- Work: maintenance coverage / takeover response
- Environment: noise / shade / green charging
- Interaction: identity-intent compliance / human alternative / exit
- Governance: near misses / complaint closure / data overreach
- Equity: satisfaction gaps for older people, children and disabled people

Geometry Registration and Drawing Precision

SUBMITTED GEOJSON + PRECISION GRADES + FIELD CALIBRATION

DIRECT GEOJSON REGISTRATION

- EPSG:4548 · drawing rotated 90° (north points right)
- Shared spine RD-001: 82-node curve · total length 9.66 km
- Closed low-speed loops RD-014/015: actual closed loops
- Green segments GRN-001/002/003: entire spine lies within the green corridor
- Corridor nodes outside all key areas: 2

PRECISION A · RECOMPUTABLE

- Site area · green-space ratio · public-space ratio · road-network length
- Directly recomputed from submitted geometry in EPSG:4548

PRECISION B · STRUCTURALLY RELIABLE

- Spine direction · three-area roles · node assignment · phasing sequence
- Structural conclusions hold; positions are indicative

PRECISION C · INDICATIVE LOCATION

- Spine alignment · section width · loop dimensions · parcel boundaries
- Uncertainty approximately ±200 m; replace with field survey

PRECISION D · NOT USABLE

- Statutory redlines · ownership · heritage-control lines · levels · utilities
- Not supplied by this proposal; official data required

P0 FIELD CHECKS BEFORE IMPLEMENTATION

- 1 Survey park centreline, ownership boundary and clear width segment by segment
- 2 Inventory accessibility breaks, slopes, tactile paving and fire access
- 3 Obtain official heritage-control lines for former Tsinghuayuan Station
- 4 Obtain rail entrances, underground utilities, levels and stormwater interfaces
- 5 Confirm land and ownership for feasible east-west stitching
- 6 Survey six HMF baselines: incidents / near misses / complaints / satisfaction gaps

CONCLUSION

- Structure and governance can inform pilot decisions; alignment, sections and scale require field verification before engineering implementation.

Human-Machine Friendly Jing-Zhang · Brand & Identity

07 / BRAND IDENTITY

CONCEPT IDENTITY SYSTEM · NOT A FORMAL VI DELIVERY

PRIMARY MARK · CONCEPT - Rail (heritage) + shared spine + station (handover) - Rust red · ink blue · leaf green · amber accent - Credit: Dr. Future wepon	NAMING SYSTEM - Primary: Human-Machine Friendly Jing-Zhang - Subtitle: Low-Speed Shared Spine - Tagline: One spine · three areas · five scenes / five dimensions · multiple nodes / two foundations	THREE POSITIONINGS TRANSLATED - Centennial Jing-Zhang Cultural Belt → rail-memory line - Urban AI-life Experience Belt → slow-mobility sharing - AI Integrated Innovation Belt → shared operation and feedback
APPLICATIONS - Wayfinding: spine signs · transfer points · H0 quiet signs - Device identity: status light · intent display · accountable-party code - Digital: twin dashboard · open-day registration · complaint portal - Events: open day · hackathon · sharing drills	PERMITTED USE - Concept communication and open-call display only; proportional scaling permitted - Retain bilingual primary name; pair with H0-H3 rights and five-dimension evaluation - Registration, trademark and commercial use require separate rights review	PROHIBITED USE - No implication of government endorsement, completed registration or exclusivity - No claims of uniform speed, construction setting-out or investment commitment - Do not remove concept / provisional notices; do not imply co-branding with undeclared third parties

International Outreach & Global Operations Map

08 / GLOBAL OPERATIONS

CONCEPT MECHANISM · AGENT.6 · NO FUNDING OR GOVERNMENT COMMITMENT

ANNUAL RHYTHM

- Q1 Foundation: publish sharing charter · train safety staff · admission drills
- Q2 Visibility: HMF Open Day · international observer visits
- Q3 Co-creation: open-source hackathon · scenario-admission assessment week
- Q4 Consolidation: developer conference · annual review and disclosure

OUTREACH → CONVERSION FUNNEL

- 01 Be seen: bilingual booklet · open repository · media observation day · short-video narrative
- 02 Arrive: international observers · university exchange · joint enterprise test week
- 03 Participate: hackathon · sharing drills · scenario admission · public co-creation
- 04 Embed: protocol reuse · standards output · team retention · next-year agenda

TAKE-AWAY ASSET KIT

- Sharing-charter template: admission / identification / exit / complaints
- Five-dimension acceptance sheet + near-miss definition
- HMF Open Day toolkit: routes / interpretation / accessibility support
- Bilingual one-pager + concept-image licensing note

BOUNDARY STATEMENT

- Events and attraction are conceptual mechanisms, not government commitments
- No assumed budget, organizer or approval result
- International communication limited to public information and open-source outputs
- Formal foreign-affairs and investment activities require separate initiation

Field Differentiation: What This Proposal Holds

COMPARISON OF FREQUENT OPEN-CALL NARRATIVES · NO NAMED CRITICISM

CORE PROPOSITION • Open Commons: open protocols and co-creation operations • Rail translation: herringbone heritage and developer pilgrimage • This proposal: human-machine-environment shared order	SPATIAL SPINE • Open Commons: narrative line / node loop • Rail translation: symbolic linear framework • This proposal: 82-node curved spine + assigned nodes	RIGHT-OF-WAY / PUBLIC VALUE • Others: limited grading or slow-mobility friendliness • This proposal: executable H0-H3 rules • Accessibility · exit · takeover · vulnerable groups
EVIDENCE HONESTY / TAKE-AWAYS • Precision grading + unmeasured baselines stated honestly • Sharing charter + acceptance sheet + P0 checklist	MAIN RISKS • Open Commons: homogenization / slogan collision • Rail translation: strong heritage but shallow institutions • This proposal: brand communication needs sustained reinforcement	

Jing-Zhang HMF Twelve Two-Hour Periods

10 / TWELVE HOURS

POINTS ON RD-001 · TIME-SHARED RIGHTS · CONCEPT OPERATIONS, NOT A STATUTORY TIMETABLE

23-05 · QUIET / MAINTENANCE

- 23-01 Zi: quiet return · H0 · Governance · SN-006 mid-corridor green spine · human patrol only; autonomous machines return to depot
- 01-03 Chou: charging/maintenance · H3 · Work · SN-010 Zhongzhiyuan edge · charging only in H3, not parks or housing
- 03-05 Yin: off-peak cleaning · H2 · Work · edges of three public layers · noise controlled; H0 prohibited

05-11 · COMMUTE / SCHOOL / WORK

- 05-07 Mao: open and yield · H1 · Movement · RD-001 · machines yield; staff open gates
- 07-09 Chen: school peak · H1 · Movement · SN-008 interchange · strict yielding or pause
- 09-11 Si: collaborative workshop · H2 · Work · SN-009 enterprise station · humans sign important decisions

11-17 · REST / TOUR / CAMPUS

- 11-13 Wu: shared meal/rest · H1 · Service · park seating / Dazhongsi · delivery ends at transfer bay
- 13-15 Wei: opt-out guided walk · H1 · Social · SN-004 / SN-006 · human interpretation in parallel
- 15-17 Shen: campus stitching · H1 · Movement · east-west university gateways · machines enter passing bays

17-23 · HOME / CARE / PATROL

- 17-19 You: homebound yielding · H1 · Movement · entire spine · machines dock or downgrade
- 19-21 Xu: evening elder support · H1 · Service · SN-007 health navigation · staffed / screen-free option
- 21-23 Hai: patrol handover · H2 · Governance · SN-003 dual verification · human-led, anonymized machine assistance

EXCLUSIONS

- No low-altitude delivery through parks · autonomous shuttle platoons · L4-L5 fully unmanned operation · uniform speed · copied island timetable
- Open/prohibited periods require transport, heritage, community and emergency approval before trial.

Three Field-Acceptance Gates

FAIL THE GATE, DOWNGRADE THE WHOLE CLASS · HUMAN ACCESS NEVER DOWNGRADES

G-01 DEPOT-RETURN CLOSURE · Zi 23:00-01:00

- After 23:00, autonomous machines in H0 / SN-006 = 0
- One machine found = FAIL that night
- FAIL 3 consecutive nights → that model remains H3-only
- Location: SN-006 · RP-02

G-02 PEAK YIELDING · Chen 07-09 / You 17-19

- Machines must not occupy the RD-001 pedestrian centreline
- Centreline occupation = 1 near miss
- Near misses ≥1 that day → prohibited same period next day
- Location: RD-001 · RP-03

G-03 EXTERNALIZED LOGISTICS · ALL DAY, ESPECIALLY Wu 11-13

- Delivery bodies inside park / H0 / seating = 0
- One machine in park = FAIL that day
- Close park-side delivery that day; retain H3 only
- Location: transfer bay · SN-010

FIVE IDENTIFIABLE COMPONENTS · REUSE EXISTING RENEWAL PROJECTS

- FC-01 Quiet-return post · Zi count · lockable park segment · SN-006 / RP-02
- FC-02 Peak passing bay · off centreline at both peaks · RD-001 / RP-03
- FC-03 Transfer bay · food stops here · Dazhongsi / RP-15
- FC-04 Emergency-stop post + screen-free window · anyone can stop / ask for help · all areas / RP-16
- FC-05 H3 charging loop · maintenance outside green spine · SN-010 / RP-16

DOWNGRADE RULE

- Failure attaches to model and time period. Restore only after depot-return / yielding / handover drill passes on site.
- Before all six P0 checks, gates are closed drills only. Repeated failure returns segment access to the previous phase.
- Human movement, accessibility and emergency access never downgrade.

Guideline Translation · Jing-Zhang Field Playbook

12 / IMPLEMENTATION PLAYBOOK

PLAN → BUILD → TEST-CERTIFY-ACCESS → OPERATE · LOCAL RULES ONLY

S1 PLANNING ASSESSMENT · Guideline 12.1

- Close the HMF impact assessment: transport / pedestrian flow / heritage / accessibility
- Do not open the spine until closed · RP-01 · P0

S2 SYNCHRONIZED CONSTRUCTION · Guideline 12.2

- Concealed works + handover drill
- Synchronized acceptance of beacons / emergency stops / transfer bays
- No night testing until drill PASS · FC-01-05 · RP-16

S3 TEST-CERTIFY-ACCESS · Guidelines 2.6 / 8.3

- Test → verify → connect
- Fencing / takeover / emergency stop / logs: any missing item blocks access
- G-01-03 · RP-05/08

S4 OPERATIONAL REVIEW · Guidelines 7.2 / 12.3

- Human-in-the-loop + rule iteration
- Redlines / human signature / 12-month revision
- Failure of any friendliness dimension blocks opening · RP-17 · RP-18

ENTERPRISE THREE-STEP

- 1 Test: Zhongzhiyuan closed loop or H3 edge · register model / task / time / responsible person
- 2 Certify: Jing-Zhang compatibility check · interface / insurance / emergency stop / safety strategy
- 3 Access: authorize by H0-H3 and G-01-03 · missing near-miss writeback = quarterly FAIL

FIVE-DIMENSION FIELD ACCEPTANCE

- Mobility: clear width / turning / five break types
- Work: operating-machine registry · maintenance only in H3
- Environment: low glare · charging outside green spine
- Interaction: intent light · staffed window · screen-free route
- Governance: collection notice · one-touch avoidance · human in the loop

NOT TRANSFERRED TO JING-ZHANG

- Low-altitude delivery · autonomous-driving-only lanes · L4-L5 fully unmanned operation
- External certification brands · first-purchase/first-use policy · fiscal subsidies

Concept Presentation Set

Generative presentation images · Precision C (indicative locations)

These images communicate spatial intent and scenario relationships. The concept WebP files remain unchanged.

Building bases, street names, section dimensions and depicted people are illustrative—not valid for measurement, setting-out, investment promotion or performance commitments.

If a concept image conflicts with an English technical figure or metrics.json, the technical figure and machine-readable data control.

C-01 Overall concept · C-02 Quantitative structure · C-03 Three key areas and shared spine

C-04 H0-H3 rights and blue-green system · C-05 Linkage monitoring framework

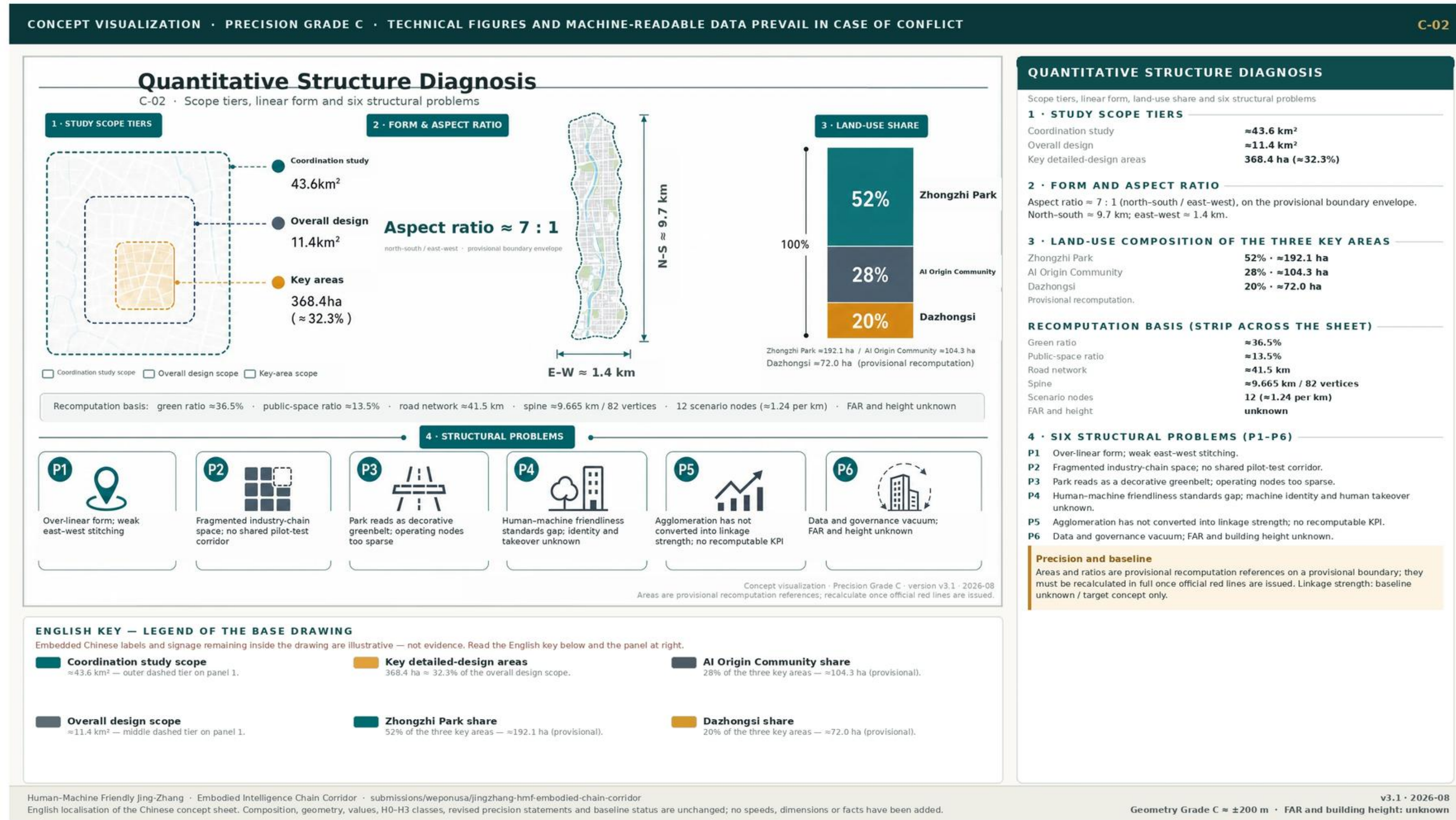
C-06 Twelve-hours scenario image · does not replace the technical clock

Illustrative concept image · Precision C · technical figures and machine-readable data control



C-02 · Quantitative Structure

Illustrative concept image · dimensions and baselines require technical / field verification



C-03 · Three Key Areas and Shared Spine

Illustrative concept image · locations are indicative and remain provisional



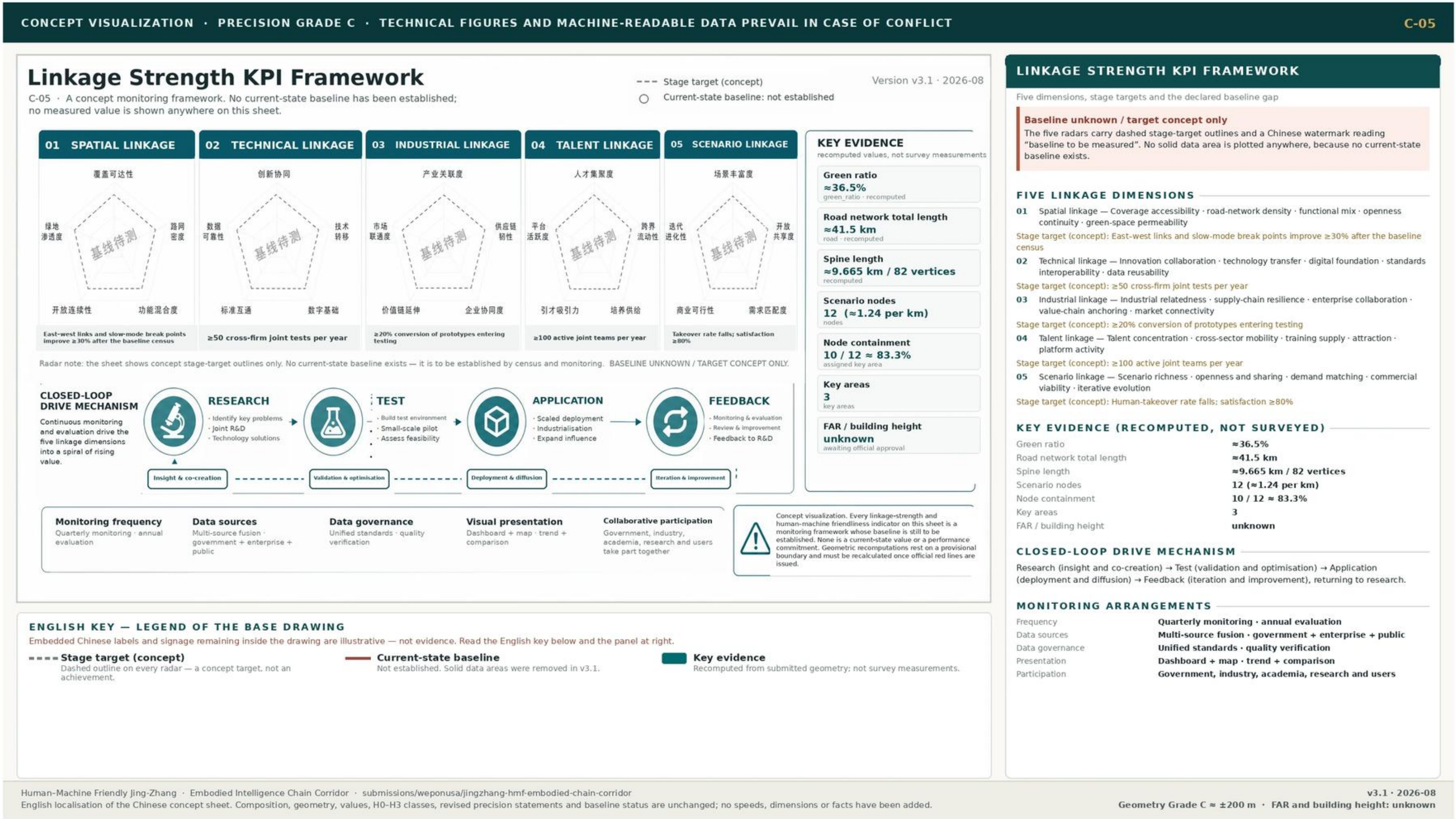
Illustrative concept image · not for measurement, setting-out or engineering

C-04

v4.0 / BILINGUAL V2

C-05 · Linkage Monitoring Framework

Current HMF baselines are unknown · no depicted target is a measured current value



C-06 · Twelve-Hours Scenario Image

Concept operations only · does not replace the English technical clock or approvals

Human-Machine Friendly Jing-Zhang · Twelve Hours of the Day

Scenario impressions for the twelve time-shared right-of-way clauses hung on spine RD-001 — park sleeps first, people take the peak, logistics stay in H3

C-06
v3.5 · wepon

CONCEPT VISUALIZATION · PRECISION GRADE C · TECHNICAL FIGURES AND MACHINE-READABLE DATA PREVAIL IN CASE OF CONFLICT



01 23:00-01:00 Quiet Return-to-Depot
H0 Human patrol only; autonomous machines must return to depot



02 01:00-03:00 Recharge & Maintenance
H3 Charging and servicing stay in H3 — not in the park or housing



03 03:00-05:00 Off-Peak Cleaning
H2 Cleaning off-peak and noise-controlled; H0 closed to machines



04 05:00-07:00 Open Gates, Give Way
H1 Commuting starts; machines yield, staffed door service



05 07:00-09:00 School & Commute Peak
H1 Strictest yielding, or machine entry suspended



06 09:00-11:00 Co-Working Studio
H2 Assistive collaboration; important decisions signed by a human



07 11:00-13:00 Shared Meals & Rest
H1 Delivery stops at the handover bay, never at seating or play areas



08 13:00-15:00 Guided Slow Walk
H1 Opt-out tour with staffed narration; heritage core stays H0



09 15:00-17:00 Campus Stitching
H1 Between classes, machines wait in pull-over bays



10 17:00-19:00 Homeward Priority
H1 Evening peak: machines dock or downgrade; people absolutely first



11 19:00-21:00 Evening Elder Support
H1 Elder navigation plus a staffed window; works without a smartphone



12 21:00-23:00 Patrol Handover
H2 Human patrol leads; machines aggregate anonymously, no face gating

Illustrative only — people, buildings, machines and street scenes are drawn impressions, not real individuals, built projects or survey data. Embedded Chinese signage inside the photographs is illustrative, not evidence.
The twelve hour-clauses are concept operating proposals; opening and no-entry periods require separate transport, heritage, community and emergency approval before any trial. Linkage strength: baseline unknown / target concept only.